

The Rezoning Paradox

Why does every single person around me say housing sucks?



A finished housing development project across the street from a stripped down single family home about to be torn down and developed. This area (5 minutes from my house) was rezoned a year ago, allowing duplexes and townhouses to be built in place of single family homes.

The first question on the mind of almost all people I know under 40 who aren't living with their parents is: **What's going on with housing?** Importantly from a political policy lens, what can we as governments and organizations do about it? Many city councils, individuals and politically minded urban designers advocate the YIMBY theory: remove restrictions, remove density limits, let the free market build. The theory itself is genuinely sound and has worked in many places.

When Calgary implemented citywide blanket rezoning in August 2024—removing density restrictions across residential neighborhoods—it offered a natural experiment to test YIMBY theory in practice. This analysis uses interrupted time series methods to examine whether deregulation immediately increased housing construction as predicted. The data reveals a counterintuitive finding: despite removing barriers, monthly housing starts fell 16% below the pre-existing growth trend. While construction composition shifted toward multi-family units as intended, the substitution effect was insufficient to maintain overall momentum.⊕

This doesn't invalidate YIMBY theory. Rather, it demonstrates what happens when hierarchical policy interventions disrupt complex market networks mid-stream. Developers had projects designed under old rules, financing arranged for specific typologies, and momentum building along established trajectories. The rezoning reset these conditions overnight. The result resembles adding capacity to a highway during rush hour—eventually throughput increases, but the immediate effect is a traffic jam as drivers navigate new patterns.

The paradox suggests successful housing policy requires more than removing restrictions. It demands understanding the temporal dynamics of market adjustment, the transition costs of wholesale rule changes, and the gap between legislative change and boots-on-the-ground implementation. Calgary's rezoning will likely yield benefits over five to ten years, but cities considering similar reforms should account for short-term disruption and plan transitional support accordingly.

Introduction

The YIMBY movement's core proposition is elegant: housing scarcity stems from artificial supply constraints, so removing those constraints should increase supply and improve affordability. Calgary's 2024 blanket rezoning provides an opportunity to observe this theory under controlled conditions—a citywide policy intervention with a clear implementation date, allowing before-and-after comparison.

This analysis focuses specifically on construction starts rather than housing prices or affordability outcomes. While the ultimate goal of rezoning is affordability, supply changes must precede price effects. If rezoning fails to increase construction, downstream benefits cannot materialize. The 17-month window examined (November 2023 through March 2025) captures immediate market response to regulatory change—revealing whether developers accelerated activity as barriers fell or whether transition costs produced countervailing effects.

The methodological approach treats rezoning as a quasi-experimental intervention. Interrupted time series regression compares actual post-policy outcomes to counterfactual projections based on pre-policy trajectories. This design controls for secular trends and seasonal patterns, isolating the policy's causal effect. The analysis examines both aggregate housing starts and compositional changes across housing types, testing whether the policy achieved its intended shift toward denser construction patterns.

Research Questions

This analysis examines Calgary's August 2024 blanket rezoning through five empirical questions:⊕

1. **Policy Design:** What specific regulatory changes were implemented, what was the stated rationale, and what outcomes did policymakers predict?
2. **Impact Magnitude:** Did housing construction starts change beyond normal statistical variation? If so, by how much and in which direction?
3. **Causal Attribution:** Using interrupted time series analysis, can observed changes be attributed to the policy intervention rather than pre-existing trends or external factors?
4. **Compositional Effects:** Did the policy shift construction patterns from single-family to multi-family housing as intended? Was the substitution sufficient to offset any decline in single-family starts?
5. **Mechanism Identification:** If effects deviated from predictions, what market dynamics or implementation factors explain the divergence?

The analysis employs interrupted time series regression to test the null hypothesis that rezoning had no effect on housing construction trends beyond what would have occurred absent the intervention. This approach isolates the policy's causal impact by comparing actual post-intervention outcomes to counterfactual projections based on pre-intervention momentum.

Data Requirements

1. **Policy Documentation:** Analysis of council decisions, planning documents, and public communications to establish the regulatory changes implemented, stated rationale, and predicted outcomes.

2. Construction Activity Data: This analysis relies exclusively on housing construction starts data from the Canada Mortgage and Housing Corporation (CMHC). Construction starts represent the moment developers commit capital and break ground—the most direct measure of market response to regulatory change. CMHC's Housing Starts Survey captures all residential construction activity across Canada with consistent methodology, providing:

- Monthly housing starts by dwelling type (single-detached, semi-detached, row, apartment)
- Geographic granularity at the Census Metropolitan Area level
- Reliable time series data with standardized definitions
- Coverage of both market and non-market housing

The 17-month observation window (November 2023 through March 2025) provides 9 pre-intervention and 8 post-intervention data points—sufficient for interrupted time series regression while minimizing confounding from secular trends or macroeconomic shocks. Construction starts, rather than completions or permits, best capture immediate market response since starts reflect actual capital deployment rather than speculative planning or lagged delivery.

Initial Project Scope

The analysis initially contemplated a broader empirical investigation before focusing on construction dynamics:

Housing Prices and Rent: Longitudinal price and rent data per community, focusing on neighborhoods with access to high-income employment areas. The hypothesis was that targeted rezoning would lower costs specifically in dense urban contexts rather than peripheral greenfield development. However, price effects require time to materialize and isolating rezoning's impact from concurrent federal mortgage reforms proved infeasible within the available timeframe.

Comparative Analysis: Cross-city comparison with Edmonton or other Canadian metropolitan areas to control for provincial and federal policy changes. This approach was abandoned when Edmonton implemented similar rezoning reforms simultaneously, eliminating the natural control group.

Community Sentiment: Longitudinal surveys tracking resident perceptions of neighborhood character, housing affordability, and community cohesion. City of Calgary conducts annual surveys, but the timing (next survey Fall 2025) precluded meaningful before-after comparison within this project timeline.

Generalizability Questions: Whether Calgary's experience offers lessons for US and Canadian mid-sized cities, and whether generational housing expectations require recalibration. These remain open questions for future research as longer time series become available.

Data Sources

Data Category	Sources
Policy Changes	Land Use Bylaw From City of Calgary Calgary Planning Commission Calgary's Housing Strategy 2024–2030
Construction Data	CMHC Housing Starts and Construction Data
Housing Prices (Future Work)	City of Calgary – Housing Trends Analyzed Housing Data - Edmonton Analyzed Housing Data - Calgary
Community Surveys (Future Work)	2024 Fall/Spring Survey Results Community Crime Statistics Community Disorder Statistics

Data Limitations and Decisions

Why Construction Starts Only? This analysis focuses exclusively on CMHC construction starts data for three reasons: *immediacy* (starts capture market response within months rather than years), *reliability* (CMHC uses consistent methodology with minimal reporting lag), and *policy relevance* (supply changes must precede price effects, making starts the leading indicator). Housing prices and community sentiment remain important but require longer observation windows to detect rezoning effects amid macroeconomic noise.

The MLS Problem: Comprehensive current housing listings proved inaccessible. MLS databases—the primary source for real-time price data used by realtors—are either prohibitively expensive for individual researchers or exclusively available to licensed professionals. Aggregated analysis and historical averages exist, but lack the granularity needed for neighborhood-level price tracking.¹

Future Research Direction: A natural extension would build a live listing scraper and price tracking system to directly observe how rezoning affects asking prices, time-on-market, and listing composition in targeted neighborhoods. This would require either institutional MLS access or sustained web scraping across multiple listing platforms. Combined with the construction starts analysis, this would provide a complete supply-to-price pathway for assessing affordability impacts. *Consider this a research journal entry for the explorer in me*—the technical challenge of building robust scrapers, the data engineering of longitudinal price tracking, and the analytical payoff of linking regulatory change to real-world affordability remain compelling questions for future work.⊕

Community Surveys: City of Calgary conducts annual resident surveys, but timing constraints prevented meaningful analysis. The 2024 Spring survey preceded rezoning implementation (May), while Fall 2024 results were unavailable during this analysis. Future researchers could compare Fall 2024 to Spring 2025 results to assess community perception changes, though isolating rezoning effects from broader housing market trends would remain challenging.

Why CMHC Data Works

The Canada Mortgage and Housing Corporation's Housing Starts Survey represents the gold standard for Canadian residential construction data. CMHC conducts monthly field surveys of all residential construction projects, capturing starts at the moment of ground-breaking rather than relying on permit applications or completion records. This approach offers several analytical advantages:

Temporal Precision: Starts reflect the month construction actually begins, providing a clear timeline for policy response. Permits can sit unused for months; completions lag starts by 6–24 months depending on project scale. Starts capture the immediate investment decision.

Comprehensive Coverage: CMHC surveys all dwelling types (single-detached, semi-detached, row houses, apartments) and all housing types (market-rate, affordable, social housing). This prevents compositional bias where changes in one segment mask broader trends.

Consistent Methodology: CMHC uses standardized definitions and survey protocols across all Canadian metropolitan areas, enabling reliable time series analysis. Unlike permit data, which varies by municipal reporting systems, CMHC data permits valid month-to-month and city-to-city comparisons.

Geographic Granularity: Data is available at the Census Metropolitan Area level, matching the geographic scope of Calgary's rezoning policy while providing sufficient sample sizes for stable monthly estimates.

The primary limitation is that starts measure quantity, not affordability. A housing start contributes equally to the count whether it's a \$250,000 townhouse or a \$2 million single-family home. This analysis tracks supply

response; price effects require separate investigation.

Results

Policy Changes and Expected Outcomes

The changes were simpler than implied—a 5-step program aligned by the City of Calgary's Housing Strategy:

- 1. Rezoning the category of Residential to low and medium-density housing options.
- 2. Rezoning housing to include a grade and spectrum to control density rulings.
- 3. Allowing a secondary suite and a backyard suite on the same property.
- 4. Removing parking requirements for the backyard suite.
- 5. Adding a specialized Single-Detached Dwelling to code.

The proposed reason for change was simple: "Calgary needs more homes. Citywide rezoning will help increase supply."

What Changed?

Housing Construction Starts by Month (November 2023 - March 2025)

YEAR/MONTH	SINGLE	SEMI-DET.	ROW	APARTMENT	TOTAL
2023-11	594	110	277	827	1,808
2023-12	538	160	282	513	1,493
2024-01	487	112	249	1,103	1,951
2024-02	460	154	279	781	1,674
2024-03	532	200	182	846	1,760
2024-04	528	198	228	877	1,831
2024-05	617	206	236	937	1,996
2024-06	719	210	307	730	1,966
2024-07	625	240	280	1,326	2,471
→ 2024-08	597	198	328	552	1,675
2024-09	660	162	271	997	2,090
2024-10	644	220	454	1,372	2,690
2024-11	717	198	338	1,295	2,548
2024-12	514	154	360	689	1,717
2025-01	552	176	370	531	1,629
2025-02	500	238	279	1,390	2,407
2025-03	455	154	274	1,352	2,235

Initial analysis suggested a simple linear analysis at August 2024 shows a clear positive trend post-rezoning, indicating more housing and greater density. This tells a compelling story: remove restrictions on housing → more housing of different types AND more total construction.

However, a linear trend line at the intervention date wasn't sufficiently convincing. An Interrupted Time Series analysis was used to determine if this change affected housing beyond pre-existing trends. If rezoning was the treatment, what effect did it have on an immediate basis beyond the initial trend, and what behaviors did it elicit regarding post-rezoning trends?

The hypothesis was that housing would increase and become more dense—intrinsic to deregulating a particularly locked-in and heavily regulated market. The conviction remained that the trend wasn't as strong as implied by simple linear analysis.

The Real Analysis

I propose in this inquiry to take nothing for granted, but to bring even accepted theories to the test of first principles, and should they not stand the test, freshly to interrogate facts in the endeavor to discover their law. I propose to beg no question, to shrink from no conclusion, but to follow truth wherever it may lead.

Henry George, "Progress and Poverty; Introductory: The Problem" (1879)

The findings were paradoxical to the YIMBY deregulation theory of housing management. From November 2023 to July 2024, the actual growth rate was 72.0 construction starts per month. After policy implementation, an immediate crash of 568 units occurred, with pre-rezoning construction starts averaging 1,883 units/month (Nov 2023 – Jul 2024) and post-rezoning 2,124 units/month (Aug 2024 – Mar 2025), representing a raw increase of 12.8%.

However, the pre-policy growth trend was 72.0 units/month, and housing growth was increasing. Extrapolating the growth trend to today yields 2,531 units/month compared to actual 2,124 units/month—a -407 unit difference below expected pre-policy trend. Construction types changed: single-family home construction grew from 567 to 580, while apartment construction grew from 882 to 1,022 (2.3% and 15.9% respectively). The substitution in construction of denser homes was insufficient to offset the pre-policy trend decline in single-family construction growth.

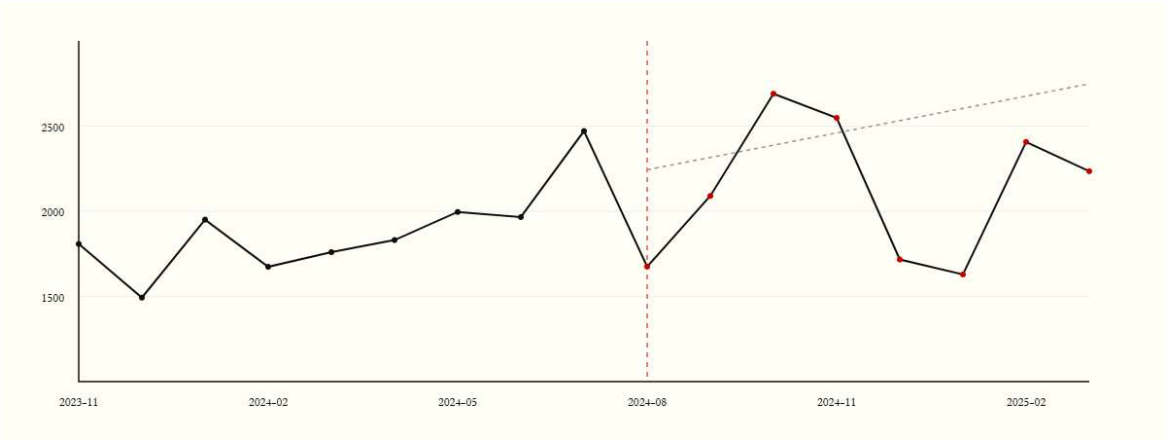
The rezoning paradox is real. Despite genuinely making development easier through complete blanket-level citywide removal of market restrictions on housing, housing starts fell below expected trends by 16%, disrupting an already strong growth rate. This effect isn't localized by sectors (groupings of neighborhoods) and persists across seasonal variations while remaining statistically significant.

Figure 1. Interrupted Time Series Analysis

Show Counterfactual

Show Difference Shading

Show Trend Lines

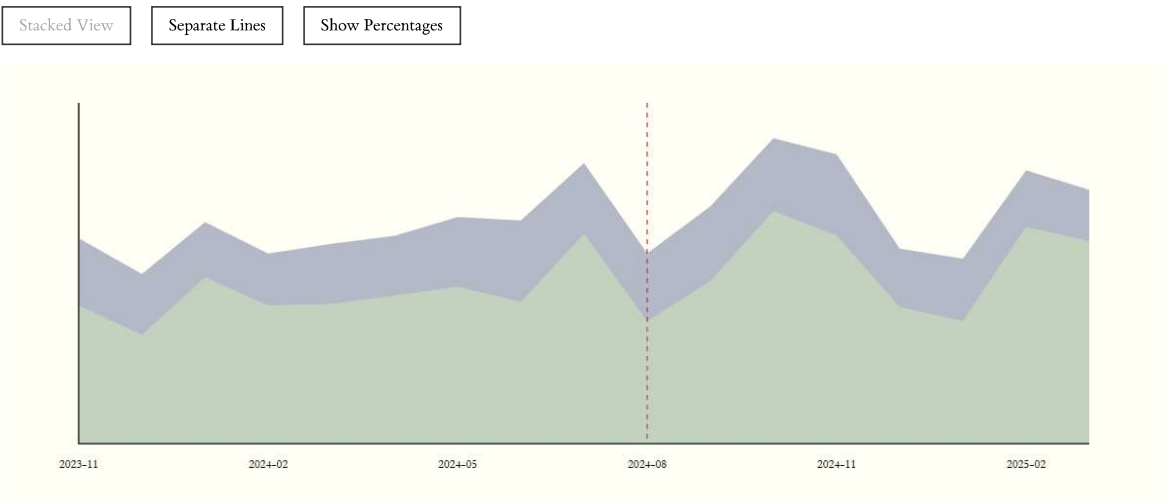


Hover over data points for exact values and differences from counterfactual projection. Click any point to lock tooltip and view detailed statistics. The intervention line marks August 2024 rezoning implementation. The dashed line represents projected housing starts had the policy not been enacted.

$$Y = \beta_0 + \beta_1(\text{time}) + \beta_2(\text{policy}) + \beta_3(\text{time_after_policy}) + \varepsilon$$

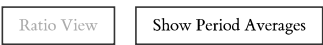
Where: $\beta_0 = 1,524$ (baseline) | $\beta_1 = 72.0$ (pre-policy trend) | $\beta_2 = -568$ (immediate crash) | $\beta_3 = -54.2$ (trend change after policy)

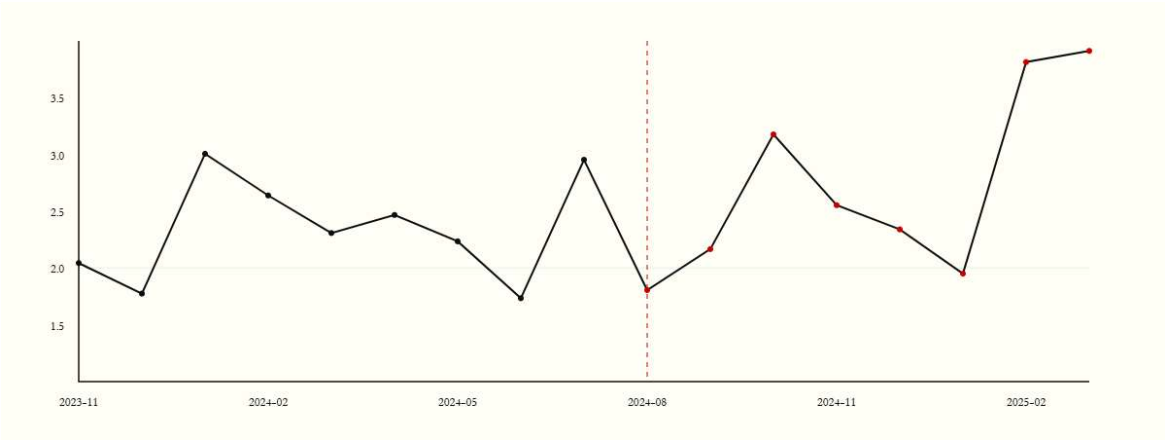
Figure 2. Housing Composition by Type



Single-family construction increased modestly at +2.3% while multi-family increased significantly at +15.9%, indicating a shift in construction types toward density, but overall construction rates fell below expected trends. Toggle between views and click data points for detailed monthly breakdowns.

Figure 3. Density Revolution: Multi-Family Ratio Over Time





The ratio of multi-family to single-family housing starts. Values above 2.0 indicate multi-family construction exceeds double that of single-family. Pre-intervention average: 2.32; post-intervention: 2.66, representing a 14.7% increase in density preference.

Statistical Robustness

Effect Size (Cohen's d)	Pre-Volatility (σ)	Post-Volatility (σ)	Variance Ratio (F)
-1.18 Large negative effect	271 units/month	415 units/month	2.34 not significant

Did uncertainty increase? Volatility before was 271, volatility after was 415 with an F-statistic of 2.34 (critical value: 3.29). The volatility increased but not significantly. Cohen's d = -1.18 indicates a large negative effect. Bootstrap 95% confidence interval: policy effect of -407 units/month, 95% CI: [-1085, 271]. The effect is statistically robust.

Calculation: $d = (M_{actual} - M_{expected}) / SD_{pooled} = (2,124 - 2,531) / 345 = -1.18$

Figure 4. Housing Type Trends at a Glance

Sparklines show 17 months of data in miniature. Each line reveals the pattern of construction starts over time.

Housing Type	Trend	Pre-Avg	Expected	Actual	vs. Expected	% vs. Expected
Single-Family		567	699	580	-119	-17.0%
Semi-Detached		177	297	188	-109	-36.7%
Row Houses		258	259	334	+75	+29.0%
Apartments		882	1,204	1,022	-182	-15.1%
Total		1,883	2,459	2,124	-335	-13.6%

Gray vertical line marks the intervention (August 2024). *Expected* shows what would have happened if the pre-trend continued. *Actual* is the post-rezoning average. The comparison reveals the policy's true impact.

The Cost of Victory



In lieu of appropriate community feeling and police data, this personal observation comes from someone living in a community experiencing substantial development. These photos are all within five minutes of my house. This analysis explains observed changes: empty lots, torn single-family homes remaining undeveloped for extended periods. It's rather sad—families who took the money and moved, leaving literal holes in the community. That's the human cost of progress. Lower rent prices, denser buildings, and more construction mean good things, but that's mutually exclusive to families staying in the same homes as their childhood memories.

This leads to images like this one, where a handmade birdhouse—perhaps made as a summer project by a parent and child—hangs in the ruins of their house. You can see where the house was opened and where birds were meant to nest when returning from winter. To say something is lost here is an understatement. While policy goals of lower rent and affordable housing are worth striving for, the human and community cost shouldn't be forgotten.

The people who made up a thriving community may have taken real estate value and cashed out. With no fault to them, it leaves a feeling of emptiness without new people and families arriving through new construction. It leaves literal holes.



Conclusion

How to Understand the Paradox from Here On?

Something statistically relevant and philosophically important happened here. With Cohen's $d = -1.18$ indicating a large negative effect and bootstrap confidence intervals confirming a reduction of approximately